

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & IT

SPRING 2025

Numerical Analysis (MS – 203)

Assignment 1

Section: A B C D & E

Semester: VI

Course Instructor: Sir Ahsan Masroor

Max Marks: 03

Question No. 1

(a) Find the positive root of the equation by Bisection Method:

$$xe^x = 1$$

which lies between $[0, 1]$

[up to 5th iteration]

(b) Use the Bisection method to find the solution for the equation:

$$2e^{-x} - \sin x = 0$$

[up to 5th iteration]

(a) Find the positive root of the equation $xe^x = 1$ in the interval $[0, 1]$ using the Bisection Method (up to 5th iteration).

Solution:

We define the function:

$$f(x) = xe^x - 1$$

We check the sign of $f(x)$ at the endpoints of $[0, 1]$:

$$f(0) = 0 \cdot e^0 - 1 = -1 \quad (\text{Negative})$$

$$f(1) = 1 \cdot e^1 - 1 \approx 2.718 - 1 = 1.718 \quad (\text{Positive})$$

Since $f(0) \cdot f(1) < 0$, a root exists in $[0, 1]$.

Bisection Iterations:

1. Iteration 1:

- Interval: $[0, 1]$
- Midpoint: $c = \frac{0+1}{2} = 0.5$
- $f(0.5) = 0.5 \cdot e^{0.5} - 1 \approx 0.824 - 1 = -0.176$ (Negative)
- New interval: $[0.5, 1]$

2. Iteration 2:

- Interval: $[0.5, 1]$
- Midpoint: $c = \frac{0.5+1}{2} = 0.75$
- $f(0.75) = 0.75 \cdot e^{0.75} - 1 \approx 1.588 - 1 = 0.588$ (Positive)
- New interval: $[0.5, 0.75]$

3. Iteration 3:

- Interval: $[0.5, 0.75]$
- Midpoint: $c = \frac{0.5+0.75}{2} = 0.625$
- $f(0.625) = 0.625 \cdot e^{0.625} - 1 \approx 1.168 - 1 = 0.168$ (Positive)
- New interval: $[0.5, 0.625]$

4. Iteration 4:

- Interval: $[0.5, 0.625]$
- Midpoint: $c = \frac{0.5+0.625}{2} = 0.5625$
- $f(0.5625) = 0.5625 \cdot e^{0.5625} - 1 \approx 0.987 - 1 = -0.013$ (Negative)
- New interval: $[0.5625, 0.625]$

5. Iteration 5:

- Interval: $[0.5625, 0.625]$
- Midpoint: $c = \frac{0.5625+0.625}{2} = 0.59375$
- $f(0.59375) = 0.59375 \cdot e^{0.59375} - 1 \approx 1.075 - 1 = 0.075$ (Positive)
- New interval: $[0.5625, 0.59375]$

Final Approximation:

The root lies in $[0.5625, 0.59375]$. Taking the midpoint:

$$\text{Root} \approx \frac{0.5625 + 0.59375}{2} = 0.578125$$

(b) Use the Bisection Method to find a solution for the equation $2e^x - \sin x = 0$ (up to 5th iteration).

Solution:

We define the function:

$$f(x) = 2e^x - \sin x$$

We first find an interval where $f(x)$ changes sign. Testing values:

$$f(0) = 2e^0 - \sin 0 = 2 \quad (\text{Positive})$$

$$f(-1) = 2e^{-1} - \sin(-1) \approx 0.736 + 0.841 \approx 1.577 \quad (\text{Positive})$$

$$f(-2) = 2e^{-2} - \sin(-2) \approx 0.271 + 0.909 \approx 1.180 \quad (\text{Positive})$$

$$f(-3) = 2e^{-3} - \sin(-3) \approx 0.100 + 0.141 \approx 0.241 \quad (\text{Positive})$$

$$f(-4) = 2e^{-4} - \sin(-4) \approx 0.037 - 0.757 \approx -0.720 \quad (\text{Negative})$$

Since $f(-4) \cdot f(-3) < 0$, a root exists in $[-4, -3]$.

Bisection Iterations:**1. Iteration 1:**

- Interval: $[-4, -3]$
- Midpoint: $c = \frac{-4+(-3)}{2} = -3.5$
- $f(-3.5) = 2e^{-3.5} - \sin(-3.5) \approx 0.060 - (-0.351) \approx -0.291$ (Negative)
- New interval: $[-3.5, -3]$

2. Iteration 2:

- Interval: $[-3.5, -3]$
- Midpoint: $c = \frac{-3.5+(-3)}{2} = -3.25$
- $f(-3.25) = 2e^{-3.25} - \sin(-3.25) \approx 0.078 - (-0.109) \approx -0.031$ (Negative)
- New interval: $[-3.25, -3]$

3. Iteration 3:

- Interval: $[-3.25, -3]$
- Midpoint: $c = \frac{-3.25+(-3)}{2} = -3.125$
- $f(-3.125) = 2e^{-3.125} - \sin(-3.125) \approx 0.088 + 0.016 \approx 0.104$ (Positive)
- New interval: $[-3.25, -3.125]$

4. Iteration 4:

- Interval: $[-3.25, -3.125]$
- Midpoint: $c = \frac{-3.25 + (-3.125)}{2} = -3.1875$
- $f(-3.1875) = 2e^{-3.1875} - \sin(-3.1875) \approx 0.083 - (-0.047) \approx 0.130$ (Positive)
- New interval: $[-3.25, -3.1875]$

5. Iteration 5:

- Interval: $[-3.25, -3.1875]$
- Midpoint: $c = \frac{-3.25 + (-3.1875)}{2} = -3.21875$
- $f(-3.21875) = 2e^{-3.21875} - \sin(-3.21875) \approx 0.080 - (-0.078) \approx 0.158$ (Positive)
- New interval: $[-3.25, -3.21875]$

Final Approximation:

The root lies in $[-3.25, -3.21875]$. Taking the midpoint:

$$\text{Root} \approx \frac{-3.25 + (-3.21875)}{2} = -3.234375$$

Question No. 2

(a) Use Newton Raphson Method to find the root of the equation:

$$x^3 - 2x - 5 = 0$$

use the initial guess of $x_0 = 2.5$

[up to 3rd iteration]

Solution:

Define the function and its derivative:

$$f(x) = x^3 - 2x - 5$$

$$f'(x) = 3x^2 - 2$$

The Newton-Raphson formula is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Iterations:**1. Iteration 1 ($x_0 = 2.5$):**

$$f(2.5) = (2.5)^3 - 2(2.5) - 5 = 5.625$$

$$f'(2.5) = 3(2.5)^2 - 2 = 16.75$$

$$x_1 = 2.5 - \frac{5.625}{16.75} \approx 2.1642$$

2. Iteration 2 ($x_1 = 2.1642$):

$$f(2.1642) \approx (2.1642)^3 - 2(2.1642) - 5 \approx 0.8106$$

$$f'(2.1642) \approx 3(2.1642)^2 - 2 \approx 12.0514$$

$$x_2 = 2.1642 - \frac{0.8106}{12.0514} \approx 2.0970$$

3. Iteration 3 ($x_2 = 2.0970$):

$$f(2.0970) \approx (2.0970)^3 - 2(2.0970) - 5 \approx 0.0265$$

$$f'(2.0970) \approx 3(2.0970)^2 - 2 \approx 11.191$$

$$x_3 = 2.0970 - \frac{0.0265}{11.191} \approx 2.0946$$

Final Approximation:

After 3 iterations, the approximate root is:

$$\text{Root} \approx 2.0946$$